Scenario 1: Image Classification Model Overfits Quickly:

You're training a CNN to classify dog breeds. You achieve 98% accuracy on training data within 5 epochs, but validation accuracy stays at 60%, and test performance is poor.

Logical Solution:
Problem: Overfitting due to a small dataset or complex model.
Fixes:
1. Add **Dropout** and **BatchNorm**
2. Use **data augmentation**
3. Apply **early stopping**
4. Use a **pretrained model** (transfer learning)

Scenario 2: ANN Fails to Learn Non-Linear Patterns

Project: Simple digit classification using **Artificial Neural Network (ANN)**
Problem: Accuracy stuck around 50%; adding more epochs doesn't improve performance.

Logical Solution:

1. ANN may not capture **non-linear features** in image data.
2. Replace ANN with **CNN** for image-based tasks.
3. Normalize input images.
4. Use **ReLU** activation instead of sigmoid/tanh

Scenario 3: CNN Model Predicts Same Class for All Images

Project: Face Mask Detection using **CNN**
Problem: Model always predicts "With Mask" for every test image.

Logical Solution:

1. Dataset is likely **imbalanced** (more "with mask" images).
2. Use **class weights** or **resample the dataset**.
3. Monitor confusion matrix and recall/precision, not just accuracy.

Scenario 4: Object Detection Model Detects Objects but Bounding Boxes Are Misaligned

Project: YOLOv8 used for Tiger Detection in Forest
Problem: Bounding boxes do not fit objects properly.

Logical Solution:

1. Labeling format might be incorrect.
2. Use correct YOLO format: (class_id, x_center, y_center, width, height) normalized to image size.
3. Check dataset with **label visualization tools** like Roboflow or CVAT.

Scenario 5: Face Mask Detection Gives High Accuracy but Performs Poorly on Real Faces

Project: Real-time Face Mask Detection using **CNN**

Problem: High accuracy on test set, but poor real-time performance on webcam feed.

Logical Solution:

1. **Training images** were likely frontal and clear; webcam has varied angles, lighting.
2. Use **data augmentation** (brightness, flip, rotation).
3. Collect **real-time webcam data** and fine-tune the model.

Scenario 6: CNN-Based Object Detector Struggles at Night

Project: YOLOv8-based object detection in CCTV
Problem: Model doesn't detect objects in dark/night images.

Logical Solution:

1. Model not trained on night/low-light images.
2. Train using **night-mode data** or **augment brightness/contrast**.
3. Apply **Histogram Equalization** or **CLAHE** preprocessing.